

Organizer Witness Vector in the Hilbert Substrate Framework: Formal Interface and Admissible Temporary Proxies

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Abstract

This note is a companion to the working organizer definition note. Its purpose is operational rather than ontological. The organizer note fixed the conceptual target: an organizer is a real mesoscale object candidate, not primary ontology, with identity given primarily by persistence of collective relief function together with enough structural and bookkeeping continuity to count as the same organizer under lawful re-expression. The present note turns that target into a formal witness interface.

The main claims are:

1. Organizer identity should be represented by a *witness vector*, not by one scalar and not by a dense reduced-state object in general.
2. The witness vector should be decomposed into functionally distinct sectors: core, shell, correlation, interface bookkeeping, bandwidth/spread, and response.
3. A lawful implementation may use temporary proxies, but those proxies must be attached to specific witness components and may not silently change the move law.
4. Dense reduced-state fidelity is demoted from general identity criterion to optional small-region auxiliary witness.
5. Retirement readiness should be expressed in the same witness language as organizer persistence, not as a separate graph-theoretic heuristic.

This note therefore provides a formal interface for future mesoscale bookkeeping work while keeping the churn rule frozen.

1 Purpose

The organizer note fixed the following principle:

Organizer identity is function-first, not graph-first.

That immediately raises the operational question:

How should organizer identity be *represented* in a lawful mesoscale implementation?

The present answer is:

By a bounded, multi-component organizer witness vector.

This note defines that vector, its sectors, its admissible temporary proxies, and its relation to no-refolding and organizer retirement.

2 Guiding principle

A correct organizer witness must satisfy all of the following:

1. It must be **local enough** to be used in candidate-move comparison.
2. It must be **rich enough** to distinguish persistent organizers from generic transient support.
3. It must be **scalable enough** to avoid dense-state blowups at larger organizer size.
4. It must be **structured enough** to separate function loss from bookkeeping loss from geometric re-expression.
5. It must be **compatible with the frozen lawful move law**, rather than serving as a pretext to change that law.

Thus the witness should not be:

- a single graph statistic,
- a single dense reduced-state fidelity,
- or an unconstrained collection of ad hoc terms.

3 Organizer witness vector

Let R be the organizer region under consideration. Define the organizer witness vector

$$\mathcal{O}_R = (C_R, S_R, M_R, K_R, B_R, Q_R). \quad (1)$$

The six sectors are:

1. C_R : **core witness sector**,
2. S_R : **shell/support witness sector**,
3. M_R : **correlation / mutual-information witness sector**,
4. K_R : **committed interface bookkeeping witness sector**,
5. B_R : **bandwidth / influence-distribution witness sector**,
6. Q_R : **collective response witness sector**.

This is a witness *vector*, not a final theorem and not a claim that every component is equally fundamental. The point is that organizer identity is multi-aspect.

4 Region of evaluation

The organizer witness vector is always evaluated in a local comparison region R . The current default region should remain the same local comparison region already adopted for the lawful move law:

1. the candidate edge or support patch,
2. immediate neighbors,
3. overlapping committed shell/core if present.

This note therefore does *not* enlarge the move law. It only refines what is measured inside the already-adopted local region.

5 Sector definitions

5.1 Core witness sector C_R

The core sector is meant to answer:

What local support center is currently carrying the dominant organizer function?

The exact definition remains open, but C_R should capture at least:

- dominant core pair or dominant core set,
- core score,
- continuity of core identity under lawful re-expression.

The crucial point is that C_R is *not* itself the whole organizer. It is only the anchor sector.

5.2 Shell/support witness sector S_R

The shell sector is meant to answer:

How is the organizer support distributed around the core, and is that support being lawfully retained or lawfully re-expressed?

S_R should capture at least:

- shell membership or shell-retention pattern,
- lawful shell weakening versus shell collapse,
- support continuity under local re-expression.

This sector is where “same organizer despite support drift” becomes operational.

5.3 Correlation witness sector M_R

The correlation sector is meant to answer:

Is the same collective relief-carrying correlation pattern still active?

M_R should capture at least:

- local mutual-information bundle,
- correlator profile,
- retained activity or equivalent local expression support,
- function-preserving versus function-destroying change.

This is the most direct witness family for function-first identity.

5.4 Committed interface bookkeeping sector K_R

The bookkeeping sector is meant to answer:

Is the same committed interface structure still doing the work?

K_R should capture at least:

- endpoint/interface bookkeeping continuity,
- interface-channel commitment,
- protected sector-specific bookkeeping when relevant.

This is the sector most directly tied to no-refolding proper.

5.5 Bandwidth / influence-distribution sector B_R

The bandwidth sector is meant to answer:

How concentrated, distributed, or slack-heavy is the organizer's influence structure?

B_R should capture at least:

- effective active-channel rank,
- influence-map rank profile,
- influence spread profile,
- slack creation versus lawful redistribution.

This sector should remain explicitly connected to finite-bandwidth and spread burden terms already present in the lawful move law.

5.6 Collective response sector Q_R

The response sector is meant to answer:

Does the same organizer respond to perturbation in the same hierarchical way?

Q_R should capture at least:

- dominant response hierarchy,
- coarse response spectrum when available,
- object-scale versus transient support-scale response patterns.

This is likely the highest-value sector scientifically, but it will often be the least available computationally. It is therefore allowed to remain sparse or absent in some implementations, provided this absence is made explicit.

6 Primary identity criterion in witness form

The organizer note adopted a function-first identity criterion. In witness-vector language, that means:

Organizer identity is preserved primarily when the *function-bearing components* of $\mathcal{O}_R$ remain sufficiently close, and only secondarily when the purely structural components remain close.

This leads to the following schematic criterion.

Let $\mathcal{O}_R^{\text{before}}$ and $\mathcal{O}_R^{\text{after}}$ be the witness vectors before and after a move. Define the organizer persistence score

$$P_{\text{org}}(M; R) = w_C P_C + w_S P_S + w_M P_M + w_K P_K + w_B P_B + w_Q P_Q, \quad (2)$$

where each $P_X \in [0, 1]$ is a sector-preservation score and the weights obey

$$w_C, w_S, w_M, w_K, w_B, w_Q \geq 0, \quad \sum_X w_X = 1. \quad (3)$$

The current conceptual requirement is:

$$w_M + w_Q + w_K \quad \text{should dominate over purely geometric weighting.} \quad (4)$$

That is the formal statement of function-first identity.

Define the organizer identity-loss witness

$$W_{\text{org}}(M; R) = 1 - P_{\text{org}}(M; R). \quad (5)$$

7 Relation to no-refolding

The organizer witness vector must fit inside the layered no-refolding program rather than replacing it.

The current layered no-refolding picture already has the form:

$$W_{\text{NR}}(M) = \alpha_{\text{ep}} W_{\text{ep}}(M) + \alpha_{\text{slack}} W_{\text{slack}}(M) + \alpha_{\text{func}} W_{\text{func}}(M) + W_{\text{sector}}(M), \quad (6)$$

where the final term is sector-specific bookkeeping loss.

The organizer witness vector should therefore enter the lawful program in the following way:

1. K_R informs endpoint/interface bookkeeping preservation,
2. B_R informs slack creation and influence redistribution,
3. M_R and Q_R inform relief-function preservation,
4. organizer-specific identity loss contributes to W_{sector} in the mesoscopic object sector.

Thus the organizer witness does *not* create a new no-refolding principle. It supplies the mesoscopic sector-specific layer already anticipated by the layered witness program.

8 Admissible temporary proxies

The witness-vector program is stronger than current implementations. So temporary proxies are unavoidable. But they must be disciplined.

8.1 General rule

A temporary proxy is admissible only if:

1. it corresponds to a named witness component,
2. it preserves the functional meaning of that component,
3. it is labeled as provisional,
4. it does not silently alter the move law.

8.2 Admissible temporary proxy families by sector

Core sector C_R . Admissible temporary proxies:

- dominant core pair,
- weighted core score,
- core continuity flag or overlap score.

Shell sector S_R . Admissible temporary proxies:

- shell retention,
- shell-edge classification (core/shell/peripheral),
- committed shell support continuity.

Correlation sector M_R . Admissible temporary proxies:

- local MI bundle,
- correlator strength profile,
- retained activity / organizer-function proxy,
- normalized local MI loss.

Bookkeeping sector K_R . Admissible temporary proxies:

- endpoint overlap witness,
- interface commitment continuity,
- sector-specific bookkeeping indicator when available.

Bandwidth sector B_R . Admissible temporary proxies:

- influence-map rank / effective-rank,
- influence spread witness,
- bounded activity plus slack profile.

Response sector Q_R . Admissible temporary proxies:

- perturbative response hierarchy,
- coarse response ordering,
- mode-participation ranking,
- explicit “not measured” marker if absent.

8.3 Demoted proxy families

The following should *not* be treated as primary organizer identity witnesses:

- raw node count,
- raw edge count,
- shell size alone,
- triangle count alone,
- graph persistence alone,
- dense reduced-state fidelity as default general identity test.

Any of these may be auxiliary diagnostics, but not the general definition of organizer identity.

9 Provisional operational interface

A bookkeeping implementation should expose the following formal outputs for each candidate move M in region R :

1. $\mathcal{O}_R^{\text{before}}$,
2. $\mathcal{O}_R^{\text{after}}$,
3. sector-preservation scores $P_C, P_S, P_M, P_K, P_B, P_Q$,
4. organizer identity-loss witness $W_{\text{org}}(M; R)$,
5. no-refolding components $W_{\text{ep}}, W_{\text{slack}}, W_{\text{func}}, W_{\text{sector}}$,
6. retirement-readiness outputs where relevant.

This is the right long-term interface because it keeps the move law fixed while making the bookkeeping inspectable.

10 Organizer witness and the frozen lawful move law

The lawful move law remains frozen in the form

$$\Delta F(M) = \Delta E_{\text{expr}} - \lambda_B \Delta C_B - \lambda_S \Delta C_S - \lambda_F \Delta C_F - \lambda_R W_{\text{NR}}. \quad (7)$$

The present note does *not* change this law. It only clarifies the bookkeeping interface:

1. ΔE_{expr} should draw primarily from the correlation/function sectors M_R and Q_R ,
2. ΔC_B and ΔC_S should draw primarily from the bandwidth sector B_R ,
3. ΔC_F should draw from organizer identity loss, especially the function-bearing parts of $\mathcal{O}_R$,
4. W_{NR} should incorporate the layered witness, with organizer identity loss contributing in the mesoscopic sector.

Thus the witness-vector note is a bookkeeping clarification of the existing law, not a new law.

11 Organizer identity-loss decomposition

To keep the bookkeeping interpretable, define

$$\Delta C_F(M; R) = \beta_{\text{id}} W_{\text{org}, \text{id}}(M; R) + \beta_{\text{func}} W_{\text{org}, \text{func}}(M; R), \quad (8)$$

where:

- $W_{\text{org}, \text{id}}$ captures structural and bookkeeping continuity loss,
- $W_{\text{org}, \text{func}}$ captures collective relief-function loss.

The present conceptual requirement is:

$$\beta_{\text{func}} \geq \beta_{\text{id}}, \quad (9)$$

because organizer identity is function-first.

This does not force function to be the only thing that matters. It only states the correct hierarchy.

12 Retirement-readiness in witness form

Organizer retirement should be expressed in the same witness language. Let v denote a candidate support patch or node. Define the retirement-readiness witness

$$W_{\text{ret}}(v; R) = \gamma_S R_S + \gamma_M R_M + \gamma_K R_K + \gamma_B R_B, \quad (10)$$

where:

- R_S : support attenuation readiness,
- R_M : low remaining function/correlation contribution,
- R_K : bookkeeping safety of removal,
- R_B : substitutability / available redistributed support.

A retirement candidate is admissible only if

$$W_{\text{ret}}(v; R) \geq \theta_{\text{ret}}. \quad (11)$$

This keeps retirement in the same organizer language rather than treating it as a graph operation.

13 Decision-status table

The note adopts the following status distinctions.

Required commitments

- organizer identity is multi-aspect,
- organizer identity is function-first,
- dense reduced-state fidelity is not the default general criterion,
- organizer retirement must be expressed in the same witness language.

Adopted current working commitments

- six-sector witness vector $\mathcal{O}_R$,
- weighted sector-preservation score P_{org} ,
- organizer identity-loss witness W_{org} ,
- admissible temporary proxy families listed above.

Open commitments

- exact final definitions of each sector,
- final sector weights,
- exact collective response descriptor,
- final retirement-readiness thresholding,
- whether different organizer species require different witness emphases.

14 What would count as success

The witness-vector program succeeds if it makes all of the following possible:

1. distinguish persistent organizers from transient support clusters,
2. identify lawful re-expression without demanding rigid graph persistence,
3. admit retirement only after functional evacuation,
4. scale to larger organizer regions without dense-state memory walls,
5. preserve the frozen lawful move law while making bookkeeping more honest.

15 What would count as failure

The companion program fails if one or more of the following occur:

1. The witness vector reduces in practice to graph descriptors only.
2. The function-bearing sectors never add information beyond raw connectivity.
3. The sector decomposition becomes so unconstrained that any behavior can be post hoc rationalized.
4. The bookkeeping still requires dense-state objects for ordinary organizer scales.
5. Retirement-readiness remains indistinguishable from raw sparsity.

16 Immediate consequences for implementation

This note implies the following immediate implementation discipline:

1. keep the churn rule frozen,
2. rewrite bookkeeping in terms of named organizer witness components,
3. log sector-preservation scores explicitly,
4. label all proxies by witness sector,
5. stop treating dense reduced-state fidelity as the default organizer identity object.

17 Conclusion

The organizer note fixed what an organizer is supposed to be. The present note fixes how organizer identity should be represented operationally.

The correct object is not a single graph statistic and not a general dense reduced-state fidelity. It is a bounded witness vector whose sectors separately track anchor structure, support re-expression, correlation/function, committed interface bookkeeping, bandwidth/spread, and collective response.

This is the right level of formality for the next stage of the mesoscape program. It preserves the frozen lawful move law, clarifies what is being measured, makes provisional proxies honest, and gives a principled route away from dense-state bookkeeping without collapsing organizer identity into arbitrary graph persistence.